

Decision tree: Crossing between rasters and vectors

st_transform() then vect()

crop() stops on a CRS mismatch, mask() warns and returns all NA, and extract() transforms with a warning: settle the CRS first

then

What does the question need?

a raster trimmed to the vector

crop(x, v)

cuts the raster to the bounding box of the vector, so the grid gets smaller; mask = TRUE also sets the cells outside the vector to NA

a raster with the outside blanked

mask(x, v)

sets the cells outside the vector to NA and leaves the grid where it was, so the extent does not change

raster values at vector locations

terra::extract()

one row per point, in the order of the points; every covered cell for a polygon, unless fun summarizes them

a vector written onto a grid

rasterize()

field sets what a cell holds, fun summarizes the features that fall in one, and background sets the rest

features from a raster

as.polygons() then st_as_sf()

dissolves the cells of each class into one multipolygon, whose boundaries are cell edges rather than real ones